

Abstract No.

Title: Functional Differences Between NSCs and GSCs in Proliferation and Quiescence

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**Background:** Glioblastoma (GB) aggressiveness and therapeutic resistance are heavily driven by glioblastoma stem cells (GSCs), a subpopulation capable of entering a reversible quiescent state. While evidence suggests GSCs arise from the malignant transformation of neural stem cells (NSCs), the precise molecular and cellular alterations governing their distinct proliferative and quiescent states remain poorly understood.

**Objective:** This study aims to compare murine NSCs and GSCs at both the cellular and gene expression levels under proliferative and quiescent conditions to uncover specific markers that distinguish healthy versus malignant stem cell states.

**Methods:** We established an in vitro model using murine NSCs and GSCs (NPE-GFP) under proliferative and serum-deprivation-induced quiescent conditions. Cell cycle state distributions were analyzed via flow cytometry. Cellular content was thoroughly evaluated using Hoechst/Pyronin Y staining to quantify DNA/RNA content and O-propargyl-puromycin (OPP) incorporation to measure nascent protein synthesis. Finally, transcriptomic profiling via RNA sequencing was integrated to dissect and compare the gene expression networks governing healthy and malignant quiescence programs.

**Results:** Our cellular analyses revealed that quiescent NSCs and GSCs successfully accumulated in the G0/G1 phase. Quiescence induction significantly reduced cellular RNA content and decreased protein synthesis in both cell types, consistent with a low metabolic state. However, comparative transcriptomic analyses and Gene Set Enrichment Analysis (GSEA) demonstrated that physiological quiescence in NSCs and malignant quiescence in GSCs are governed by distinct gene expression programs, showing minimal overlap in their respective signatures and highlights.

**Conclusion:** Although NSCs and GSCs share common cellular features during quiescence, such as cell cycle arrest and metabolic depression, they activate fundamentally different regulatory pathways to maintain their stemness and state. These unique transcriptomic differences offer valuable insights into the basic biology of GB quiescence, paving the way for the identification of tumor-specific biomarkers and therapeutic targets that spare healthy NSC function.



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